5.3.1 Tangent Lines & Arc Lengths of Polar Curves

Tangent Lines of Polar Curves

Motivation: Consider an arbitrary polar curve C of the form $r = f(\theta)$. Suppose we want to find the equation of a tangent line to some point on the curve.

Theorem: In order to find the slope of a tangent line on a polar curve at a given angle θ we calculate:

Example 1: For the cardioid $r = 1 + \sin(\theta)$ find the slope of the tangent line at the point $\left(1 + \frac{\sqrt{3}}{2}, \frac{\pi}{3}\right)$.

Example 2: Find the points on the cardioid $r = 1 + \sin(\theta)$, on the interval $[0, 2\pi)$ where there are horizontal & vertical tangent lines and singular points.

Example 3 (you try): Find the points on the curve given by $r = 1 - \cos(\theta)$ from $0 \le \theta < 2\pi$ where there are horizontal & vertical tangent lines and singular points.

<u>Example 4:</u> The three-petal rose $r = \sin(3\theta) + 1$ has three tangent lines at the origin. Find the polar equation which represents each of these lines.

<u>Arc Length of Polar Curves</u>

Motivation: We now consider the arc length of a polar curve.

Formula: The length of a curve with polar equation $r = f(\theta)$ where $a \le \theta \le b$, is

Example 5 (you try):

(a) Find the length of one tracing of the cardioid given by $r = 1 + \sin(\theta)$. You may use a calculator to approximate the value of the definite integral.

(b) How would the problem change if you were working with function $r = \sin(\theta)$?

Example 6 (you try):

Find the *exact* length of the cardioid given by $r = e^{\theta}$ where $0 \leq \theta \leq \pi$. You may use a calculator to approximate the value of the definite integral.